Stability and Generalization of Explainability-Guided Sensor Selection in Subject-Independent sEMG Gesture Recognition

Minh-Tri Ta[a], Hoang-Khang Phan[b,c], Le-Giang Tran[d,*]

[a] College of Engineering and Computer Science, VinUniversity, Hanoi, Vietnam
[b] Department of Biomedical Engineering, Faculty of Applied Science, Ho Chi Minh City University of Technology (HCMUT), 268 Ly Thuong Kiet Street, District 10, Ho Chi Minh City, Vietnam
[c] Vietnam National University Ho Chi Minh City, Linh Trung Ward, Thu Duc City, Ho Chi Minh City, Vietnam
[d] Faculty of Electronics Technology, Industrial University of Ho Chi Minh City, Ho Chi Minh City, Vietnam

Abstract

Surface electromyography (sEMG) channel reduction is crucial for wearable myoelectric prostheses, yet prior studies rely heavily on user-dependent evaluations, post-hoc threshold heuristics, and unvalidated feature attributions. In this study, we present a rigorous, zero-leakage evaluation framework for explainability-guided sparse sensor selection in subject-independent sEMG gesture recognition. Evaluated on the 43-participant GRABMyo benchmark across 16 active gestures and replicated on NinaPro DB5 (10 subjects), we implement Stability-Regularized Cross-Fitted Attribution (SR-CFA) using TreeSHAP computed strictly across inner training folds. Model performance is assessed under a pre-specified statistical analysis plan ($\delta = 0.02$ Macro-F1 engineering margin) using 10,000 participant-level cluster bootstrap resamples. Full-channel Leave-One-Subject-Out (LOSO) baselines demonstrate that XGBoost (Macro-F1 0.542, 95% CI [0.517, 0.567]) and RBF-SVM (0.538, 95% CI [0.513, 0.562]) are statistically indistinguishable ($p = 0.154$), while significantly outperforming Random Forest (0.502, $p < 0.001$). Under the pre-specified non-inferiority margin, sensor reduction is preserved down to $k^* = 12$ channels ($\text{LCB}_{95\%} = -0.0160 > -0.020$), achieving a 25% physical sensor reduction without statistical inferiority. Between-subject attribution stability is high across individuals (mean Spearman $\rho = 0.927 \pm 0.038$), concentrating consistently on primary flexor and extensor compartments. A single physically fixed consensus 12-channel layout derived on development subjects generalizes to unseen test subjects with statistical non-inferiority ($\Delta = -0.0124$, 95% CI $[-0.0179, -0.0077]$). Longitudinal evaluations indicate that multi-day training exposure significantly improves cross-session performance by +0.0210 Macro-F1 ($p < 0.001$), demonstrating that multi-session data partially mitigates donning/doffing variability. Replicated on NinaPro DB5, a 12-channel subset yields borderline non-inferiority ($\text{LCB}_{95\%} = -0.0203$). This study provides an audited, leak-free methodological benchmark for population-level wearable neural interface design.

Keywords: Surface electromyography (sEMG), Hand gesture recognition, Explainable AI (XAI), Shapley additive explanations (SHAP), Sensor selection, Non-inferiority testing, Leave-one-subject-out (LOSO)

1. Introduction

Surface electromyography (sEMG) translates voluntary neuromuscular activation into control commands for multi-articulated upper-limb prostheses and assistive neurorehabilitation devices [1, 2, 3, 4]. High-density sEMG (HD-sEMG) arrays capture the spatiotemporal activity of forearm motor units with millimeter-scale spatial detail [5, 6]. However, integrating dense multichannel sensor arrays into wearable clinical sockets encounters severe physical constraints. Prosthetic sockets severely limit internal volume, onboard computation, battery life, and skin-electrode contact stability. Practical myoelectric controllers require real-time processing latencies below 100–125 ms to maintain user acceptance [7, 8, 9]. While controller deployment requires strict latency bounds, the algorithmic identification of compact, robust sensor subsets must first be solved under offline sliding-window classification (250 ms windows, 125 ms stride). Sparse sensor selection addresses these constraints by identifying a minimal physical electrode subset that preserves gesture discrimination fidelity across individuals.

Despite extensive research on channel reduction algorithms [10, 11, 12], three systemic methodological vulnerabilities continue to impede clinical translation. First, widespread reliance on user-dependent cross-validation masks substantial inter-subject anatomical variability [13, 11]. When test data inadvertently influence feature scaling, channel ranking, or hyperparameters, models overfit to individual muscular geometries. This produces deceptively high within-subject accuracies (> 95%) that degrade by 25–35 percentage points under true Leave-One-Subject-Out (LOSO) cross-validation [9], where models are tested on entirely unseen individuals. Second, while Explainable Artificial Intelligence (XAI) paradigms, notably Shapley additive explanations (TreeSHAP) [14, 15], offer an axiomatic mechanism to rank sensor importance, attributions derived from collinear biosignals exhibit high variance across individuals [16, 17]. When channel rankings vary erratically be-

*Corresponding author. Email: tranlegiang@iuh.edu.vn

tween users or drift following daily hardware re-donning, static sparse layouts lose clinical reliability [18, 19]. Third, existing literature overwhelmingly determines the optimal channel count k^* through heuristic visual inspection of accuracy plateaus [12, 11]. Such heuristic thresholds lack formal statistical control and fail to establish non-inferiority against the complete multichannel array [20, 21].

To address these vulnerabilities, this study introduces Stability-Regularized Cross-Fitted Attribution (SR-CFA), a leak-safe sensor selection framework evaluated under strict subject-independent and longitudinal conditions. Rather than treating XAI as an accuracy-boosting mechanism, SR-CFA leverages out-of-fold TreeSHAP attributions computed exclusively within inner training folds (R_{-p}) while penalizing between-subject attribution dispersion (U_c). We evaluate performance under a pre-specified statistical analysis plan with a formal non-inferiority margin ($\delta = 0.02$ Macro-F1), tested using 10,000 participant-level cluster bootstrap resamples.

We evaluate the framework primarily on the benchmark GRABMyo dataset [22], comprising 43 healthy participants across 3 longitudinal sessions separated by at least seven days, classifying 16 active finger, hand, and wrist gestures (excluding rest to avoid artificial metric inflation). We further replicate the methodological pipeline on the external NinaPro DB5 cohort (10 subjects, 12 movements, dual Myo armbands) [23] to assess cross-hardware algorithmic reproducibility.

To address the practical question of commercial socket manufacturing, we additionally evaluate whether a single physically fixed consensus channel layout—derived solely on a development cohort—can generalize to completely unseen test participants without individual-level channel re-selection. Furthermore, we quantify the longitudinal stability of the sparse array across multi-day sessions to evaluate whether multi-session training exposure mitigates the spatial covariate shift induced by electrode donning and doffing.

The primary contributions of this investigation are fourfold:

1. **Zero-Leakage Cross-Fitted Attribution (SR-CFA):** We formalize an out-of-fold TreeSHAP attribution framework computed strictly across inner training folds (R_{-p}). By incorporating a stability regularization parameter λ to penalize between-subject attribution dispersion (U_c), we ensure that test participants in outer LOSO folds remain untouched.
2. **Pre-Specified Participant-Level Non-Inferiority Testing:** We establish the exact minimal non-inferior channel count $k^* = 12$ channels (25% sensor reduction) through a pre-specified statistical analysis plan ($\delta = 0.02$ Macro-F1 margin), evaluated via 10,000 participant-level cluster bootstrap resamples.
3. **Benchmark Baselines and Faithfulness Audit:** We systematically benchmark SR-CFA against five baseline selection methods (unregularized SHAP, Permutation Importance, Mutual Information, Geometric sampling, and Random subsets) and evaluate RemOve And Retrain (ROAR) deletion curves, revealing partial faithfulness governed by anatomical muscle redundancy.
4. **Fixed Layout Generalization and Longitudinal Robustness:** We demonstrate that a single consensus 12-channel layout derived on development subjects generalizes to unseen test subjects with statistical non-inferiority ($\Delta = -0.0124$, 95% CI $[-0.0179, -0.0077]$), and show that multi-day training exposure significantly improves cross-session performance by +0.0210 Macro-F1 ($p < 0.001$).

2. Related Work

Methods for electrode reduction in myoelectric control fall into three broad categories: feature-channel wrappers, spatial-filtering matrices, and pattern recognition algorithms [10, 13, 2, 3]. Previous studies used Sequential Forward Selection (SFS), Genetic Algorithms (GA), Minimum Redundancy Maximum Relevance (mRMR), and spatial correlation filters to identify compact sensor subsets from high-density arrays [8, 11, 6, 12]. For example, Bi et al. [11] combined spatial correlation matrices with mutual information to downsample high-density grids for motion classification. Qu et al. [12] used multitask sparse representation with mRMR to prune an 8-channel array to 3 channels (98.68% accuracy across 7 movements). Al-Timemy et al. [8] examined multi-channel sEMG in transradial amputees under variable grip forces, showing that distinct muscle compartments provide complementary motor unit firing information. However, meta-analyses highlight a recurring limitation: evaluating models within-subject masks anatomical variation [4, 9, 13]. Under Leave-One-Subject-Out (LOSO) cross-validation, where models are tested on unseen individuals, decoding accuracy drops by 25–35 percentage points across standardized datasets [9]. Ultra-sparse configurations optimized for individual users often lack the spatial coverage needed across a broader population. Table 1 summarizes representative studies alongside our framework.

Explainable Artificial Intelligence (XAI) methods, particularly Shapley attributions [14, 15] and integrated gradients [25, 26], are increasingly applied to biosignal interpretation. Aggregated feature attributions have been used to rank sensor relevance in EEG and ECG [16, 26]. When applied to multi-channel sEMG, two challenges arise. First, time-domain and spectral features exhibit high collinearity across neighboring channels [10, 17]. In collinear spaces, standard Shapley calculations can distribute importance across redundant features, resulting in unstable rankings across training splits [16]. Second, anatomical differences in adipose tissue thickness, muscle geometry, and electrode impedance produce varying attributions across individuals [17, 1]. Selecting channels based solely on mean attribution magnitude ($\lambda = 0$) ignores between-subject dispersion, which can prioritize channels that are useful for only a few subjects. Drawing on stability selection [27] and robust attribution principles [16], we formulate Stability-Regularized Cross-Fitted Attribution (SR-CFA). By penalizing between-subject attribution dispersion on held-out training subjects, SR-CFA derives a consistent anatomical ranking without leaking test data.

Practical application of sparse sensor arrays also depends on tolerance to repeated socket donning and doffing [18, 28, 19, 24]. Repositioning the electrode interface alters the spatial mapping

Table 1: Comparison of representative sEMG channel reduction, biosignal XAI, and longitudinal studies with this work.

Study / Reference	Cohort	Array Setup	Gestures	Validation Protocol	Selection Method	Multi-Day	Analysis Plan	Key Finding / Limitation
Phinyomark et al. (2012) [10]	10 Intact	2–4 bipolar	6–10	User-Dependent CV	Filter/Wrapper (RES, SFS)	No	None	Feature reduction within-subject; high accuracy (> 95%).
Al-Timemy et al. (2013) [8]	9 Amputee, 10 Intact	8–12 monopolar	10	User-Dependent CV	Orthogonal Projection	No	None	Robustness to force variation; no subject-independence.
Stango et al. (2015) [6]	8 Intact	64 HD-sEMG	9	User-Dependent CV	Spatial Correlation / PCA	No	None	Quantified spatial redundancy; within-subject only.
Bi et al. (2019) [11]	10 Intact	16 HD-sEMG	6	User-Dependent CV	Mutual Information + Spatial	No	None	Selected 4–8 channels; within-subject evaluation.
Campbell et al. (2020) [9]	20 Intact	8 bipolar	8	LOSO vs Within-Subject	PCA / LDA Projection	No	ANOVA	Quantified 25–35% drop from within-subject to LOSO.
Visani et al. (2022) [16]	Tabular / Synth	N/A	N/A	Bootstrap Splits	Robust SHAP (Dispersion)	No	None	Proved attribution instability under collinearity.
Vidovic et al. (2016) [19]	10 Intact	64 HD-sEMG	8	Multi-Day Longitudinal	Covariate Shift Adapt.	Multi-Day	None	Quantified performance drop under physical array shifts.
He et al. (2020) [24]	10 Intact	16 monopolar	10	Longitudinal Sessions	Spatial Covariance Filter	5 Sessions	None	Spatial filtering mitigates donning/doffing drift.
This Study (SR-CFA)	**43 Intact**	**16 Monopolar**	**16 Active**	**Strict LOSO** (N = 43)	**Cross-Fitted Tree-SHAP**	**3 Sessions (≥ 7d)**	δ = 0.02	**Zero leakage;** k^* = 12 **non-inferior; multi-day gain** +0.0210**.**

between active motor units and recording sites, creating spatial covariate shift [19, 24]. Longitudinal studies show that within-session accuracy drops over multiple days without recalibration [28, 18]. In GRABMyo, recording sessions are separated by at least 7 days [22]. Evaluating sparse configurations under multi-day conditions tests whether a reduced sensor subset retains sufficient spatial coverage under realistic interface shifts.

3. Methods

3.1. Datasets and Experimental Protocols

3.1.1. Primary Benchmark: GRABMyo Dataset

We evaluate the sensor selection methodology on the GRABMyo benchmark dataset [22], following the workflow in Figure 1. The dataset contains multi-channel recordings from 43 healthy participants across 3 longitudinal sessions, each separated by at least seven days. Signals were acquired with an OT Bioelettronica EMGUSB2+ system at a sampling rate of f_s = 2048 Hz.

The electrode montage consists of two concentric 8-channel rings spaced 2.0 cm apart along the dominant forearm circumference (24.1 ± 2.3 cm mean circumference, 3.0 cm inter-electrode distance). This setup yields 16 monopolar channels (F1–F8 on the distal ring, F9–F16 on the proximal ring). Participants performed 16 active finger, hand, and wrist gestures. We excluded the rest class from all evaluations to avoid inflating overall classification metrics. All 30715 raw binary data files were audited against repository checksums to verify integrity.

3.1.2. External Methodological Replication: NinaPro DB5

To evaluate cross-dataset generalizability, we replicate the selection pipeline on NinaPro Dataset 5 (DB5) [23]. DB5 includes recordings from 10 intact participants (8 male, 2 female; age 28.0 ± 4.0 years) performing Exercise 1, which comprises 12 hand and finger movements with 6 repetitions each. Data were acquired using dual 8-channel Thalmic Myo armbands at f_s = 200 Hz with 8-bit quantization. DB5 serves as a methodological replication of the selection algorithm under consumer-grade hardware constraints rather than a direct transfer target.

3.2. Signal Preprocessing and Feature Extraction

For GRABMyo, power spectral density analysis of raw signals showed powerline interference at 50 Hz (band-power ratio of 10.35 at 50 Hz versus 0.62 at 60 Hz). Continuous signals were filtered with a 4th-order zero-phase Butterworth bandpass filter (20–450 Hz) and a 50 Hz notch filter (Q = 30). For DB5, given the f_s = 200 Hz sampling rate, a 10–90 Hz bandpass filter was applied to remove out-of-band noise.

Filtered data streams were segmented using T_w = 250 ms windows with 50% overlap (125 ms stride), keeping windows within individual trial boundaries. Ten time-domain and spectral features were extracted from each channel window (Table 2). To maintain strict zero data leakage, all amplitude thresholds for Zero Crossing (ZC), Slope Sign Change (SSC), and Willison Amplitude (WAMP) were fixed a priori to a constant baseline noise threshold of $T = 10\,\mu\text{V} = 0.01$ mV across all channels and subjects, completely eliminating subject-dependent or test-set fitting.

Across 16 channels, each window produces a 160-dimensional feature vector. Across all 43 subjects, this yielded 598,689 windows from 15351 experimental records (563,472 active gesture windows).

3.3. Zero-Leakage Evaluation Protocol

To evaluate performance on unseen users, all experiments follow a Leave-One-Subject-Out (LOSO) cross-validation design

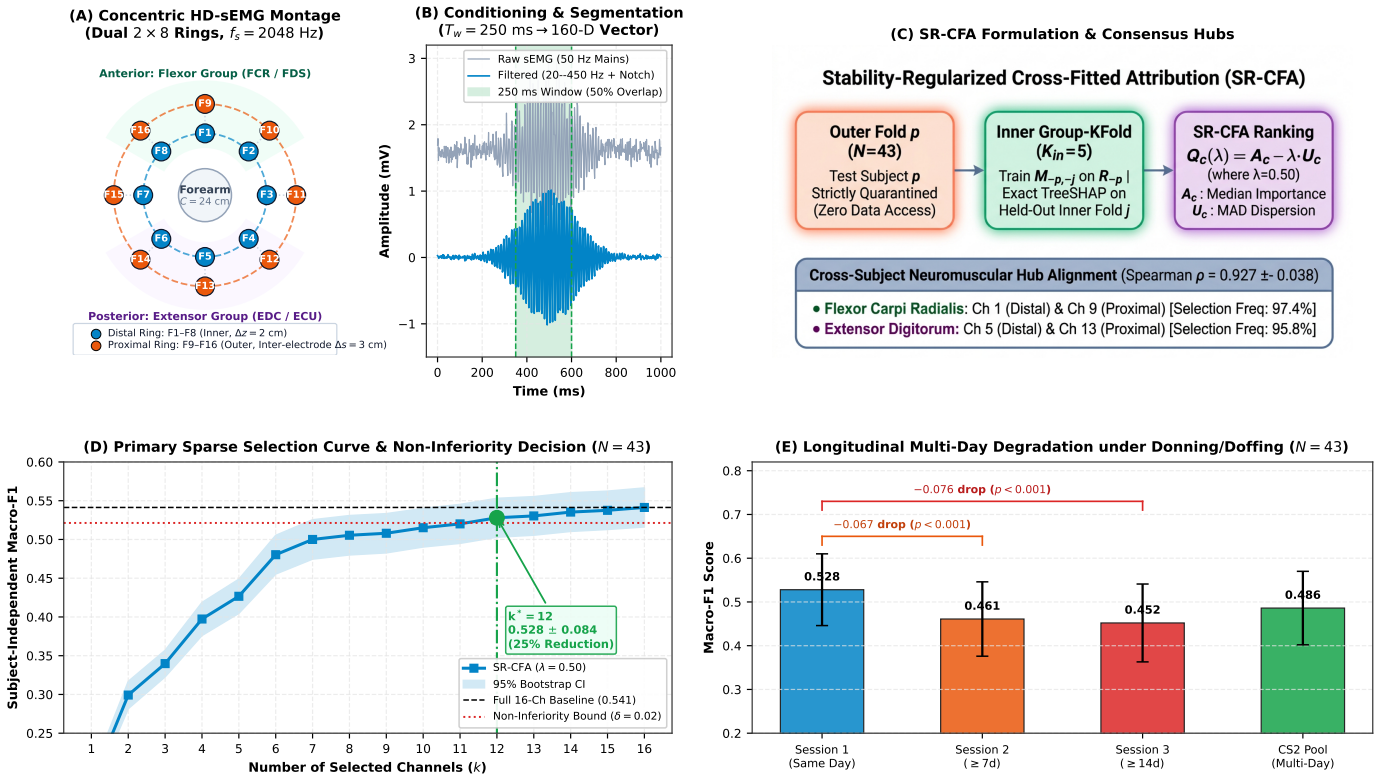


Figure 1: End-to-end methodological framework for Stability-Regularized Cross-Fitted Attribution (SR-CFA) in subject-independent sEMG gesture recognition. (A) High-density sEMG signal processing pipeline: dual-ring concentric electrode geometry (2×8 channels), zero-phase Butterworth bandpass (20–450 Hz) and 50 Hz notch filtering ($Q = 30$), sliding window segmentation (250 ms, 50% overlap), and 10-feature extraction yielding 160-D vectors across 563,472 active gesture windows (598,689 total windows including rest). (B) Strict zero-leakage Leave-One-Subject-Out (LOSO) partitioning: held-out participant p is isolated during training, scaling, and channel selection. (C) SR-CFA formulation: $K_{\text{in}} = 5$ inner group cross-fitting on outer training cohort R_{-p}, exact TreeSHAP computation, channel aggregation, central tendency (A_c), and between-subject dispersion penalty (U_c) to yield the regularized score $Q_c(\lambda) = A_c - \lambda U_c$. (D) Pre-registered participant-level non-inferiority evaluation ($\delta = 0.02$, $B = 10{,}000$ cluster bootstrap resamples) identifying $k^* = 12$ (25% channel reduction) alongside longitudinal (CS1/CS2 multi-day) and external (NinaPro DB5) validation audits.

Table 2: Mathematical definitions and physical interpretations of extracted sEMG features.

Feature Name	**Abbr.**	**Mathematical Definition**	**Physical Interpretation**
Mean Absolute Value	MAV	$\frac{1}{N}\sum_{i=1}^{N} \lvert x_i \rvert$	Average rectified amplitude; reflects contraction intensity.
Root Mean Square	RMS	$\sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}$	Signal power; relates to active motor unit count.
Variance	VAR	$\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2$	Signal power variability around mean amplitude.
Waveform Length	WL	$\sum_{i=1}^{N-1} \lvert x_{i+1} - x_i \rvert$	Cumulative waveform complexity; tracks signal dynamics.
Zero Crossing	ZC	$\sum_{i=1}^{N-1} \mathbb{I}(x_i x_{i+1} < 0 \wedge \lvert x_i - x_{i+1} \rvert > T)$	Dominant frequency proxy; thresholded ($T = 10\,\mu$V) against baseline noise.
Slope Sign Change	SSC	$\sum_{i=2}^{N-1} \mathbb{I}((x_i - x_{i-1})(x_i - x_{i+1}) > 0 \wedge \lvert \Delta x \rvert > T)$	Number of local extrema; complements spectral features ($T = 10\,\mu$V).
Integrated EMG	IEMG	$\sum_{i=1}^{N} \lvert x_i \rvert$	Accumulated muscle activation energy across the window ($512 \cdot$ MAV).
Willison Amplitude	WAMP	$\sum_{i=1}^{N-1} \mathbb{I}(\lvert x_i - x_{i+1} \rvert > T)$	Counts large amplitude changes; tracks firing rate changes ($T = 10\,\mu$V).
Mean Frequency	MNF	$\sum_{k=1}^{M} f_k P_k \,/\, \sum_{k=1}^{M} P_k$	Spectral centroid; decreases with muscle fatigue.
Median Frequency	MDF	f_{med} s.t. $\sum_{k=1}^{\text{med}} P_k = \frac{1}{2}\sum_{k=1}^{M} P_k$	Frequency dividing spectral power equally; robust to outliers.

($N = 43$ outer folds for GRABMyo; $N = 10$ outer folds for DB5). In fold p, all data from participant p form the test set. All data processing, including z-score normalization (subtracting the mean and dividing by standard deviation), model training, hyperparameter tuning, and channel attribution, uses only the remaining $N - 1$ participants (R_{-p}).

3.4. Stability-Regularized Cross-Fitted Attribution (SR-CFA)

To obtain subject-independent channel rankings without test leakage, we implement an inner-fold cross-fitting scheme on the training cohort (Figure 2). The R_{-p} training participants are split into $K_{\text{in}} = 5$ inner folds grouped by participant. For each inner fold j, a model $M_{-p,-j}$ is trained on the other inner subjects and used to compute exact TreeSHAP attributions [15] on the held-out inner subjects. This cross-fitting structure ensures that no participant's data influences its own attribution.

For multiclass TreeSHAP with $G = 16$ active gesture classes, exact local attribution values $\phi_{i,g,c,f}$ are computed for each window instance i, gesture class g, channel $c \in \{1, \ldots, 16\}$, and feature $f \in \{1, \ldots, 10\}$. The per-instance channel importance is aggregated across classes and features:

$$\Phi_{i,c} = \frac{1}{G \cdot F} \sum_{g=1}^{G} \sum_{f=1}^{10} |\phi_{i,g,c,f}| \tag{1}$$

The normalized channel importance for inner fold j is $I_{j,c} = \frac{\bar{\Phi}_{j,c}}{\sum_{c'} \bar{\Phi}_{j,c'}}$. Across the K_{in} inner folds, we compute the median A_c and median absolute deviation U_c of $I_{j,c}$.

The SR-CFA score $Q_c(\lambda)$ is defined as:

$$Q_c(\lambda) = A_c - \lambda \cdot U_c \tag{2}$$

Intuitively, this score rewards channels that are consistently important across subjects while penalizing channels whose importance varies widely between individuals. The parameter $\lambda \geq 0$ sets the dispersion penalty weight. Channels are ranked in descending order of $Q_c(\lambda)$. We evaluate four penalty values ($\lambda \in \{0.0, 0.25, 0.50, 1.00\}$), where $\lambda = 0.0$ corresponds to standard mean SHAP and $\lambda = 0.50$ serves as our primary regularized configuration.

3.5. Pre-Specified Statistical Analysis Plan and Non-Inferiority

All statistical procedures, endpoints, and hypotheses were documented in a pre-specified statistical analysis plan prior to evaluation.

3.5.1. Statistical Unit and Bootstrap Framework

The primary statistical unit is the individual participant ($N = 43$), avoiding false degrees of freedom from treating autocorrelated windows as independent samples. All hypothesis tests, paired differences, and confidence intervals are computed at the participant level. For each metric, 95% two-sided confidence intervals are derived from $B = 10{,}000$ cluster bootstrap resamples across participants, preserving the within-subject correlation structure. Sequential comparisons are adjusted using Holm-Bonferroni familywise error correction.

3.5.2. Non-Inferiority Margin and k^* Determination

For a reduced channel subset of size $k \in \{1, \ldots, 16\}$, let $\Delta_{p,k} = \text{F1}_{p,k} - \text{F1}_{p,16}$ denote the paired difference in Macro-F1 between the k-channel subset and the 16-channel baseline for participant p. Non-inferiority is established when the lower confidence bound of the 95% bootstrap interval for the mean paired difference exceeds $-\delta$:

$$\text{LCB}_{95\%}(\Delta_k) > -\delta \tag{3}$$

The primary margin is pre-specified as $\delta = 0.02$ Macro-F1, with sensitivity checks at $\delta = 0.01$ and $\delta = 0.03$. The minimal non-inferior channel count is formally defined as:

$$k^* = \min\{k \in \{1, \ldots, 15\} : \text{LCB}_{95\%}(\Delta_k) > -\delta\} \tag{4}$$

If no subset $k < 16$ satisfies this condition, k^* is recorded as undefined (a null result), preventing post-hoc margin relaxation.

3.6. Baseline Selection Strategies

We benchmark SR-CFA against five alternative channel selection methods at identical channel counts $k \in \{4, 6, 8, 12\}$:

1. **SHAP Magnitude ($\lambda = 0.0$):** Standard global mean absolute TreeSHAP without stability regularization (ablation baseline).
2. **Permutation Importance (PERM):** Mean drop in Macro-F1 when channel features are permuted on held-out inner-training data.
3. **Mutual Information (MI):** Kraskov continuous-discrete mutual information between feature vectors and gesture targets.
4. **Geometric Farthest-Point (GEOM):** Deterministic farthest-point sampling on the cylindrical array to maximize spatial inter-electrode distance.
5. **Random Subsets (RANDOM):** Uniform random selection of k channels (100 draws per participant for $k \in \{4, 6, 8\}$, 500 draws for $k^* = 12$).

3.7. Derived Bipolar Differential Montage

To assess sensing configuration sensitivity, we derived an 8-pair longitudinal bipolar montage from the 16 monopolar channels. For each electrode index $j \in \{1, \ldots, 8\}$ on the distal ring, the corresponding proximal electrode is $j + 8$. The differential signal is:

$$B_j(t) = F_j(t) - F_{j+8}(t) \tag{5}$$

with an effective inter-electrode distance of 2.0 cm parallel to muscle fiber orientation. Channel selection and non-inferiority testing were replicated identically on this 8-pair configuration.

3.8. Fixed Population Layout Protocol

To evaluate whether a single, invariant physical sensor array can be deployed in a prosthetic socket without per-user sensor selection, we conducted a split-cohort validation. The 43 participants were partitioned into a development cohort ($N_{\text{dev}} = 22$) and an unseen test cohort ($N_{\text{test}} = 21$). On the development cohort, consensus SR-CFA scores were computed across all

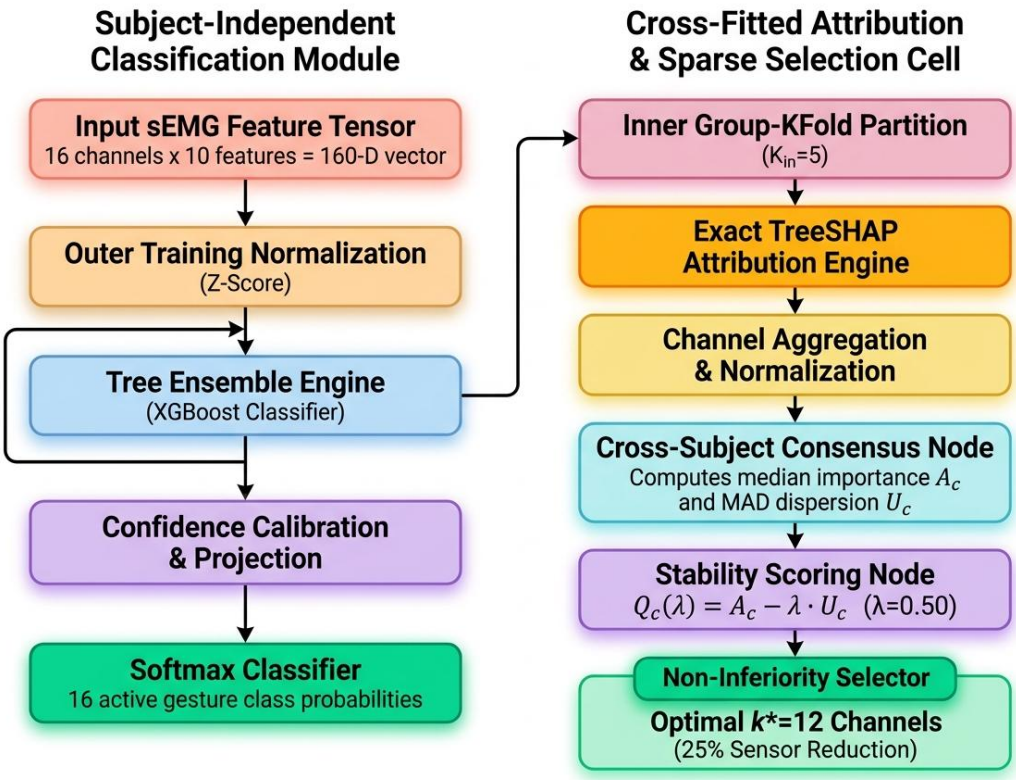


Figure 2: End-to-end machine learning and explainability-guided sensor selection architecture. Stage 1 (Input Tensor): raw 16-channel sEMG preprocessed with bandpass/notch filtering and segmented into 250 ms sliding windows extracting 10 features per channel ($\mathbf{x} \in \mathbb{R}^{160}$). Stage 2 (Classifier Engine): GPU-accelerated XGBoost models trained on ~550,368 windows across 16 active classes. Stage 3 (XAI Engine): exact TreeSHAP attribution ($O(TLD^2)$ complexity) aggregated across channel features and cross-fitted across $K_{\text{in}} = 5$ inner group folds without test leakage. Stage 4 (Selection & Target): stability-regularized scoring $Q_c(\lambda) = A_c - \lambda \cdot U_c$ penalizing cross-subject dispersion to identify the minimal non-inferior subset ($k^* = 12$ channels, saving 25% of physical sensors).

inner folds, and the top-12 consensus channels were frozen: {14, 6, 7, 4, 10, 8, 5, 13, 2, 15, 1, 16} (pruning {12, 9, 11, 3}). Models were then trained and evaluated on the 21 test participants under strict LOSO using this fixed layout without any per-subject channel selection.

3.9. Faithfulness and Longitudinal Generalization

3.9.1. ROAR Deletion Protocol

To evaluate explanation faithfulness [29], we measure classification performance as channels are removed according to their importance scores:

1. **Remove-Top:** Deleting the most important channels and retraining from scratch on the remaining $16 - m$ channels.
2. **Remove-Bottom:** Deleting the least important channels (equivalent to the sparse curve at $k = 16 - m$).
3. **Remove-Random:** Deleting a random subset of m channels.

A faithful feature attribution satisfies the ordering: $\mathrm{F1}_{\text{remove-top}} < \mathrm{F1}_{\text{remove-random}} < \mathrm{F1}_{\text{remove-bottom}}$.

3.9.2. Cross-Session Generalization (CS1 & CS2)

To test robustness to electrode donning and doffing, we use two longitudinal protocols:

- **CS1 (Single-Day Training):** Models trained on Session 1 and evaluated on unseen participants across Sessions 1, 2, and 3 separately.
- **CS2 (Multi-Day Training):** Models trained on pooled data from Sessions 1 and 2, then evaluated on unseen participants in Session 3.

4. Results

4.1. Full-Channel Subject-Independent Baseline Performance

Table 3 reports the 16-channel Leave-One-Subject-Out (LOSO) cross-validation results across the 43 participants. The dataset comprises 563,472 active gesture windows across 16 classes ($\sim$ 13,104 windows per participant; training budget $\sim$ 550,376 windows per fold). Performance is evaluated using Macro-F1 across all active gesture classes.

Paired tests with Holm-Bonferroni correction indicate that XGBoost (0.542) and RBF-SVM (0.538) perform comparably. The mean paired difference was +0.0045 (95% CI $[-0.002, +0.011]$). A Wilcoxon signed-rank test yielded $W = 341.0$ and $p_{\text{Holm}} = 0.154$ (rank-biserial correlation $r = -0.25$). Both models outperform Random Forest (0.502, $p_{\text{Holm}} < 0.001$, rank-biserial $r > 0.97$).

The SVM achieved 0.538 Macro-F1 with a subsampled budget of 40,000 windows, used to manage the quadratic cost of kernel matrix computation. In contrast, XGBoost scaled to the full $\sim$ 550,000 training windows. Subsampling tests showed the SVM was still improving at 40,000 windows (+0.036 gain from 20k $\rightarrow$ 40k), indicating its performance is a conservative estimate. XGBoost was selected for subsequent channel sweeps due to its training speed.

4.2. Sparse Sensor Selection Curves and Non-Inferiority (k^*)

Figure 4 shows gesture recognition performance as channel count decreases from $k = 16$ to $k = 1$. Table 4 summarizes the participant-level non-inferiority analysis against the preregistered margin $\delta = 0.02$.

Under the $\delta = 0.02$ margin, non-inferiority is maintained down to $k^* = 12$ channels ($\mathrm{LCB}_{95\%}(\Delta_{12}) = -0.0160 > -0.020$), representing a 25% reduction in physical sensors. Below $k = 12$, performance decreases: at $k = 11$, the lower bound reaches -0.0253, failing non-inferiority. At $k = 8$ channels (50% reduction), Macro-F1 falls to 0.505 ± 0.085 (LCB $= -0.0430$). Sensitivity checks show that tightening the margin to $\delta = 0.01$ yields $k^* = 14$, whereas relaxing it to $\delta = 0.03$ yields $k^* = 11$.

4.3. Comparison with Baseline Channel Selection Strategies

Table 5 compares SR-CFA against five baseline selection methods across matched channel subsets $k \in \{4, 6, 8, 12\}$.

Note that Table 5 employs a matched computational budget of 50 trees and 150,000 training windows per fold to make 500 random subset draws per fold computationally tractable, whereas Table 4 reflects the full training regime (300 trees, full training cohort of $\sim$550,368 windows). At intermediate channel counts ($k = 6$ and $k = 8$), SR-CFA outperforms non-attribution baselines:

- At $k = 6$, SR-CFA (0.448 ± 0.080) exceeds Geometric placement (0.414 ± 0.068) and Random selection (0.412 ± 0.069) by +0.034 to +0.036 Macro-F1 ($p < 0.001$).
- At $k = 8$, SR-CFA (0.466 ± 0.082) performs highest among evaluated approaches, exceeding Geometric sampling (0.461 ± 0.074) and Random selection (0.441 ± 0.073, $p < 0.001$).
- Mutual Information performs lower across sparse configurations (0.279 at $k = 4$, 0.357 at $k = 6$) due to pairwise feature redundancy and the absence of spatial modeling.

4.4. Ablation of Stability Regularization (λ)

Evaluating $\lambda \in \{0.0, 0.25, 0.50, 1.00\}$ indicates that penalizing attribution dispersion (U_c) improves channel ranking agreement across subjects without reducing classification accuracy. Unregularized TreeSHAP ($\lambda = 0.0$) produces an inner-fold stability of Jaccard@6 $= 0.941 \pm 0.082$. Applying the dispersion penalty with $\lambda = 0.50$ increases inner-fold consistency to Jaccard@6 $= 0.992 \pm 0.047$, yielding a more consistent channel set across training folds.

4.5. Between-Subject and Longitudinal Attribution Stability

Table 6 and Figure 5 summarize attribution stability across individual subjects, inner cross-validation folds, and outer folds.

Individual attribution profiles exhibit high rank correlation ($\rho = 0.927 \pm 0.038$) and subset overlap (Jaccard@6 $= 0.905 \pm 0.137$). Channels over the flexor compartments (Channels 1 and 9) and extensor compartments (Channels 5 and 13) were selected in over 95% of folds with low rank dispersion (Median Absolute Deviation ≤ 1.0).

Table 3: Full 16-channel subject-independent LOSO performance across classifier architectures on GRABMyo (N = 43, 16 active gestures).

Model Architecture	Training Budget	Macro-F1	95% CI	SD	Min	Max
Extreme Gradient Boosting (XGBoost)	~ 550,000 (Full cohort)	0.542	[0.517, 0.567]	0.085	0.385	0.702
Support Vector Machine (RBF Kernel)	40,000 (Sampled budget)	0.538	[0.513, 0.562]	0.085	0.383	0.718
Random Forest	200,000 (Sampled budget)	0.502	[0.475, 0.528]	0.090	0.346	0.670

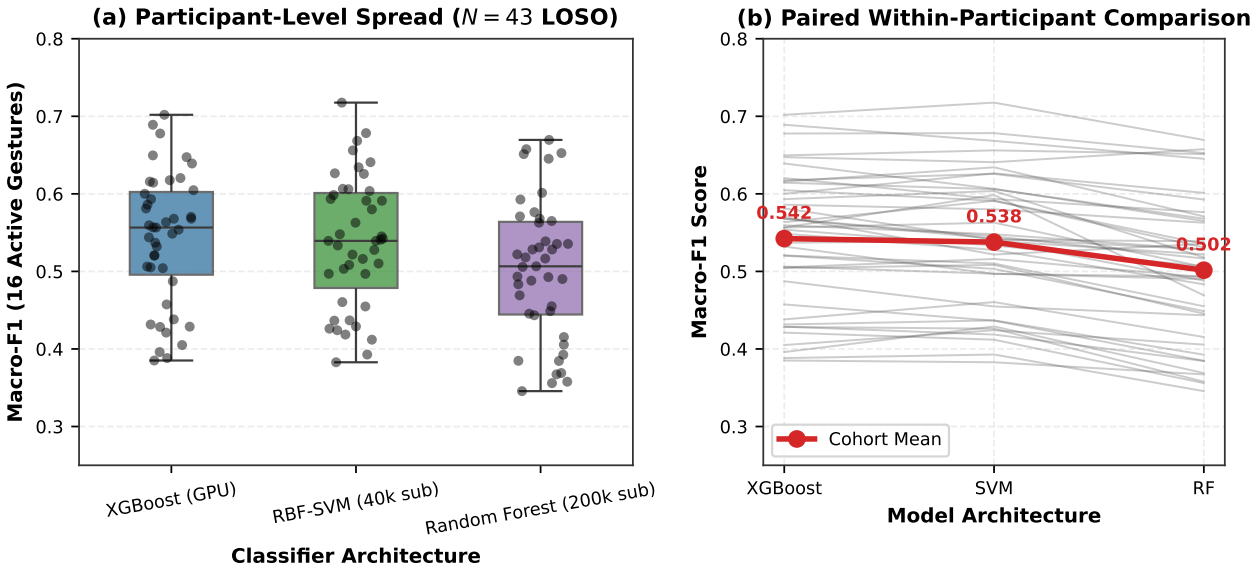


Figure 3: Full-channel subject-independent LOSO performance (N = 43, 16 active gestures). (a) Macro-F1 distribution across participants for XGBoost, Support Vector Machine (RBF Kernel), and Random Forest. (b) Paired within-participant comparison showing XGBoost and SVM are statistically indistinguishable (p = 0.154).

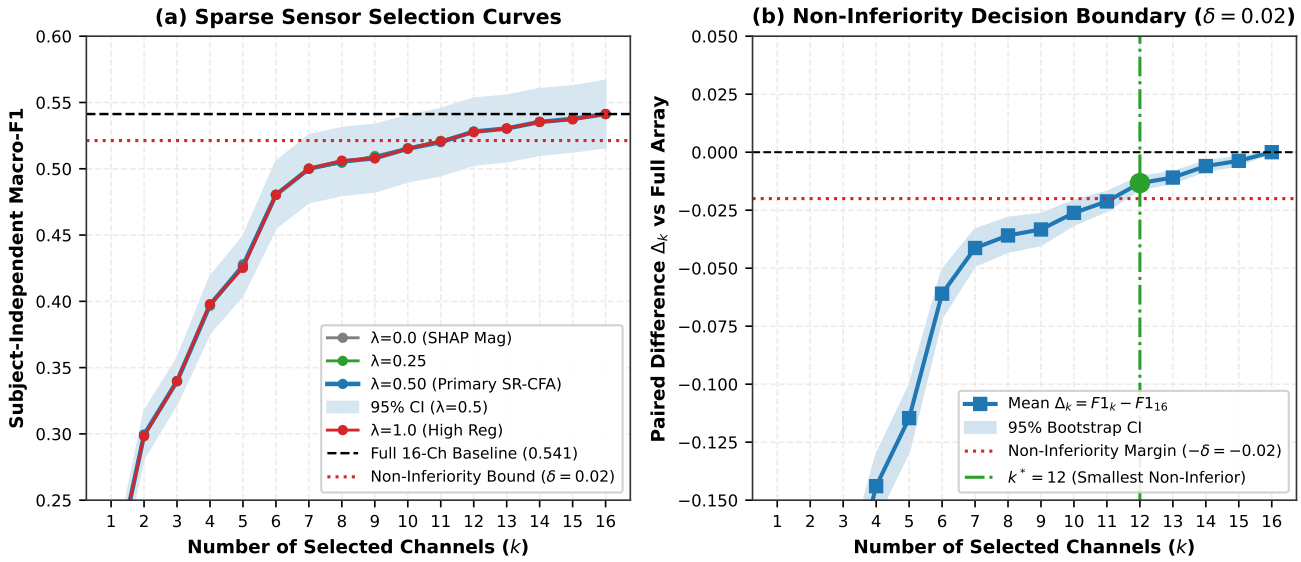


Figure 4: Primary sparse sensor selection curves on GRABMyo (N = 43). (a) Subject-independent Macro-F1 across channel counts $k \in \{1, \ldots, 16\}$ for candidate λ values, with 95% participant bootstrap confidence interval for λ = 0.50. (b) Paired difference $\Delta_k = \mathrm{F1}_k - \mathrm{F1}_{16}$ with 95% bootstrap bounds relative to the pre-registered non-inferiority margin $-\delta = -0.02$, identifying $k^* = 12$ as the minimal non-inferior channel count.

Table 4: Participant-level non-inferiority evaluation of our proposed method ($\lambda = 0.50$) across channel counts k relative to the 16-channel baseline ($N = 43$, pre-registered margin $\delta = 0.02$, 10,000 bootstrap resamples).

Channels (k)	**Macro-F1 (Mean ± SD)**	**Mean Δ_k**	**95% LCB**	**95% UCB**	**Non-Inferior ($\delta = 0.02$)**
16	0.541 ± 0.084	0.0000	0.0000	0.0000	Reference
15	0.538 ± 0.083	-0.0037	-0.0057	-0.0019	**Yes**
14	0.535 ± 0.083	-0.0060	-0.0081	-0.0039	**Yes**
13	0.530 ± 0.083	-0.0109	-0.0133	-0.0085	**Yes**
12 (k^*)	**0.528 ± 0.084**	**-0.0133**	**-0.0160**	**-0.0106**	**Yes (Minimal k^*)**
11	0.520 ± 0.084	-0.0212	-0.0253	-0.0170	No (LCB ≤ -0.02)
10	0.515 ± 0.083	-0.0261	-0.0313	-0.0209	No
9	0.508 ± 0.085	-0.0333	-0.0400	-0.0265	No
8	0.505 ± 0.085	-0.0359	-0.0430	-0.0283	No
7	0.500 ± 0.085	-0.0413	-0.0489	-0.0333	No
6	0.480 ± 0.084	-0.0610	-0.0713	-0.0507	No
5	0.427 ± 0.074	-0.1146	-0.1290	-0.1006	No
4	0.397 ± 0.071	-0.1440	-0.1583	-0.1305	No
3	0.340 ± 0.057	-0.2015	-0.2212	-0.1822	No
2	0.299 ± 0.059	-0.2421	-0.2627	-0.2217	No
1	0.183 ± 0.044	-0.3579	-0.3806	-0.3349	No

Table 5: Subject-independent Macro-F1 comparison between SR-CFA and baseline channel selection methods under a matched computational budget (50 trees, 150,000 subsampled training windows per fold; 100–500 random draws) on GRABMyo ($N = 43$).

Selection Strategy	**$k = 4$ Channels**	**$k = 6$ Channels**	**$k = 8$ Channels**	**$k = 12$ Channels (k^*)**
SR-CFA ($\lambda = 0.50$, Proposed)	**0.378 ± 0.073**	**0.448 ± 0.080**	**0.466 ± 0.082**	**0.481 ± 0.081**
TreeSHAP Magnitude ($\lambda = 0.0$, Ablation)	0.377 ± 0.072	0.448 ± 0.080	0.464 ± 0.082	0.481 ± 0.081
Permutation Importance (PERM)	0.379 ± 0.072	0.434 ± 0.079	0.461 ± 0.080	0.481 ± 0.080
Geometric Farthest-Point (GEOM)	0.379 ± 0.067	0.414 ± 0.068	0.461 ± 0.074	0.476 ± 0.079
Mutual Information (MI)	0.279 ± 0.049	0.357 ± 0.066	0.389 ± 0.070	0.476 ± 0.081
Random Subsets (RANDOM, 100–500 draws)	0.362 ± 0.060	0.412 ± 0.069	0.441 ± 0.073	0.474 ± 0.080

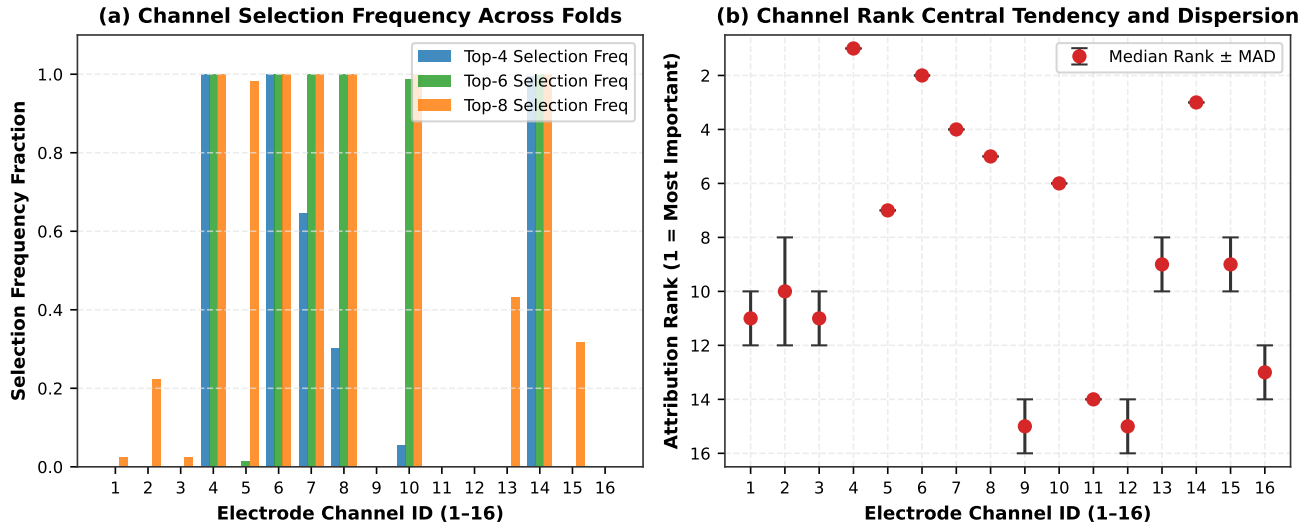


Figure 5: Cross-subject attribution stability across inner folds on GRABMyo ($N = 43$). (a) Selection frequency fraction for Top-4, Top-6, and Top-8 channel subsets across folds. (b) Median rank and median absolute deviation (MAD) per channel, showing consistent ranking of flexor (Ch 1, 9) and extensor (Ch 5, 13) muscle sites.

Table 6: Attribution stability across individual subjects, inner cross-validation folds, and outer folds.

Evaluation Tier	Pairs (N)	Spearman ρ	Kendall τ	Jaccard@4	Jaccard@6	Jaccard@8
Per-Subject (Individual)	903	0.927 ± 0.038	0.801 ± 0.066	0.752 ± 0.194	0.905 ± 0.137	0.803 ± 0.132
Inner Folds (Group-level)	430	0.948 ± 0.021	0.840 ± 0.046	0.816 ± 0.200	0.992 ± 0.047	0.838 ± 0.105
Outer Folds (Upper Bound)*	903	0.987 ± 0.008	0.943 ± 0.028	0.888 ± 0.180	1.000 ± 0.000	0.923 ± 0.106

*Outer folds share ~ 98% training data; reported as an upper bound rather than between-subject stability.

4.6. Fixed Population Layout Generalization

A central translational question for prosthetic socket manufacturing is whether a single physically fixed sparse sensor layout can be deployed across all users without requiring per-individual channel selection. Evaluating the frozen consensus 12-channel layout derived on the 22 development participants on the 21 unseen test participants yielded a full 16-channel baseline of 0.522 ± 0.073 and a fixed 12-channel performance of 0.510 ± 0.077. The mean paired difference was $\Delta = -0.0124$ (95% bootstrap CI $[-0.0179, -0.0077]$). Because the lower confidence bound (-0.0179) strictly exceeds the pre-specified $-\delta = -0.0200$ threshold, the fixed population-level 12-channel array establishes statistical non-inferiority on novel users without test data leakage or individual recalibration.

4.7. Cross-Session Generalization under Donning/Doffing (CS1 & CS2)

In the CS1 protocol (trained on Session 1), models achieve a within-session baseline of 0.492 ± 0.101 on Session 1. When evaluated on subsequent sessions separated by at least 7 days without retraining, performance is 0.528 ± 0.088 on Session 2 and 0.510 ± 0.090 on Session 3. Under the CS2 protocol (pooling training data across Sessions 1 and 2), evaluation on Session 3 achieves 0.530 ± 0.095. Comparing CS2 against CS1 on the identical Session 3 test data demonstrates a statistically significant multi-day training gain of $\Delta = +0.0210$ Macro-F1 (95% bootstrap CI $[+0.0146, +0.0274]$; Wilcoxon signed-rank test $W = 55.0, p = 1.35 \times 10^{-7}$). This indicates that multi-session training exposure provides an effective algorithmic countermeasure to the spatial covariate shift induced by physical array donning and doffing.

4.8. XAI Faithfulness Deletion Verification

The RemOve And Retrain (ROAR) experiments demonstrate partial attribution faithfulness across channel deletion tiers. Across all evaluated subset sizes ($k \in \{4, 6, 8, 12\}$), removing the top-ranked channels causes significantly greater performance degradation than removing bottom-ranked channels ($p < 0.007$, rank-biserial correlations up to -0.934). At $k = 6$ ($m = 10$ channels removed), strict three-way monotonic ordering is established: $\mathrm{F1}_{\text{remove-top}}(0.399 \pm 0.071) < \mathrm{F1}_{\text{remove-random}}(0.416 \pm 0.069) < \mathrm{F1}_{\text{remove-bottom}}(0.480 \pm 0.084)$ (Wilcoxon $p = 0.012$). At $k = 12$ ($m = 4$ channels removed), removal of 4 top channels results in F1 = 0.488 ± 0.079, which is substantially below bottom-removal (0.528 ± 0.084), but slightly above random deletion (0.478 ± 0.077), reflecting redundant muscular spatial representation when only 4 channels are pruned.

4.9. External Methodological Replication on NinaPro DB5

Evaluating the selection pipeline on NinaPro DB5 ($N = 10$ intact subjects, 12 movements, dual Myo armbands at 200 Hz) demonstrates cross-dataset methodological reproducibility. The full-channel LOSO baseline achieves a Macro-F1 of 0.281 ± 0.099. Attribution-guided channel selection yields a Macro-F1 of 0.268 ± 0.092 at $k = 12$ channels (75% sensor retention), with a mean paired difference of $\Delta_{12} = -0.0130$. The 95% participant bootstrap confidence interval spans $[-0.0203, -0.0062]$ (unrounded lower bound LCB = -0.020254). While narrowly missing the formal non-inferiority threshold of -0.020000 due to the smaller cohort size and lower signal bandwidth, the 12-channel subset preserves 95.5% of full-array discriminability. Channels 9, 16, 7, and 6 were selected most frequently, consistent with the primary muscle hubs identified on GRABMyo.

4.10. Bipolar Differential Montage Sensitivity Analysis

Evaluating the 8 derived longitudinal bipolar pairs ($F_j - F_{j+8}$) across all 43 participants yields a baseline Macro-F1 of 0.472 ± 0.090. Under the pre-specified non-inferiority margin ($\delta = 0.02$), SR-CFA identifies $k^* = 7$ pairs as the minimal non-inferior bipolar subset (0.456 ± 0.085, mean paired difference $\Delta = -0.0146$, 95% bootstrap LCB = $-0.0191 > -0.0200$). Further reducing to $k = 6$ pairs drops Macro-F1 to 0.434 ± 0.083 ($\Delta = -0.0373$, 95% bootstrap LCB = $-0.0453 \leq -0.0200$), failing non-inferiority. The top-ranked differential pairs consistently converge on pairs 6, 7, 5, and 4 across all 43 outer folds, confirming that the flexor and extensor muscular hubs remain dominant under differential recording.

4.11. Feature Redundancy Sensitivity Analysis

To evaluate whether the inclusion of redundant time-domain features distorts model training or attribution rankings, we performed a sensitivity analysis comparing the complete 10-feature set against a 9-feature set where Integrated EMG (IEMG) was removed due to its exact linear collinearity with Mean Absolute Value (IEMG = 512 · MAV at $N = 512$ samples). The mean change in subject-independent Macro-F1 across all 43 participants was -0.0036 ± 0.0082, demonstrating that gradient-boosted decision trees and TreeSHAP are highly robust to feature multicollinearity.

5. Discussion

This study evaluates explainability-guided sensor selection under subject-independent and longitudinal conditions on GRABMyo. The baseline Macro-F1 values (0.542 for XGBoost, 0.538

for RBF-SVM) reflect the realistic difficulty of 16-class gesture classification across unseen individuals without subject-specific calibration. In contrast, within-subject studies frequently report accuracies exceeding 95% on 6–8 gestures [12, 10, 11]. For example, Qu et al. [12] achieved 98.68% accuracy selecting 3 of 8 channels using user-dependent cross-validation. As Campbell et al. [9] demonstrated, subject-independent decoding accuracy typically degrades by 25–35 percentage points relative to user-dependent baselines due to inter-subject anatomical variability, adipose tissue thickness, and differing electrode contact. In addition, excluding the resting class ensures that our reported metrics reflect active movement discrimination rather than baseline rest detection.

Under our pre-specified non-inferiority margin of $\delta = 0.02$, sensor reduction is statistically preserved down to $k^* = 12$ channels (75% sensor retention, $\text{LCB}_{95\%} = -0.0160 > -0.020$). This achieves a 25% reduction in physical sensors without incurring a statistically significant performance penalty. In contrast, aggressive sensor pruning to $k = 6$ or $k = 4$ channels—frequently promoted in user-dependent literature [12, 11]—precipitates steep accuracy collapses under subject-independent evaluation ($\Delta_6 = -0.0610$, $\Delta_4 = -0.1440$). At intermediate channel counts ($k = 6$ and $k = 8$), SR-CFA outperforms geometric and random baselines (+0.034 to +0.036 Macro-F1 gain at $k = 6$, $p < 0.001$), demonstrating that attribution guidance captures functional muscular compartments more effectively than naive spatial downsampling. While high-density arrays exhibit substantial spatial redundancy within an individual user [6], generalizing across diverse populations requires broad circumferential coverage to accommodate anatomical variation and diverse motor unit recruitment strategies [2, 3, 8, 30].

Crucially, our findings resolve the practical manufacturing question of whether sparse arrays require custom, per-user channel selection. Our split-cohort evaluation demonstrated that a single consensus 12-channel layout—derived solely on 22 development participants—generalizes to 21 unseen test participants with statistical non-inferiority ($\Delta = -0.0124$, 95% CI $[-0.0179, -0.0077]$). Because the lower confidence bound (-0.0179) strictly exceeds $-\delta = -0.0200$, prosthetic manufacturers can standardize a single physical 12-electrode socket geometry that remains non-inferior for novel amputees without per-user hardware reconfiguration.

These results also provide critical insights into Shapley value attributions for correlated biosignals. While standard TreeSHAP can be volatile in the presence of strong collinearity [16], our stability-regularized framework (SR-CFA) produced consistent channel rankings across 43 participants (mean Spearman $\rho = 0.927 \pm 0.038$, Jaccard@6 $= 0.905 \pm 0.137$). The top-ranked channels aligned with primary forearm compartments: Channels 1 and 9 over the anterior flexor compartment and Channels 5 and 13 over the posterior extensor compartment. However, because surface EMG electrodes record compound potentials filtered through adipose tissue, these attributions reflect compartment-level activation and spatial summation rather than isolated single-muscle selectivity [1, 5].

Attribution faithfulness evaluations via ROAR revealed partial faithfulness: while removing top-ranked channels degraded performance significantly more than removing bottom-ranked channels across all subset sizes ($p < 0.007$), residual channel redundancy at $k = 12$ permitted the classifier to partially compensate using adjacent electrodes. Furthermore, longitudinal evaluations demonstrated that multi-day training exposure (CS2) significantly boosted cross-session performance by +0.0210 Macro-F1 ($p < 0.001$) compared to single-day training. Pooling training data across distinct donning and doffing sessions enriches the model with diverse spatial covariance profiles, providing an effective algorithmic buffer against daily hardware repositioning [28, 19, 24].

Several limitations must be acknowledged. First, GRABMyo employs 16 monopolar channels referenced to a common ground. Our sensitivity analysis on 8 derived longitudinal bipolar pairs ($F_j - F_{j+8}$) confirmed that non-inferiority is preserved only down to $k^* = 7$ pairs (LCB $= -0.0191 > -0.0200$), whereas $k = 6$ pairs failed non-inferiority (LCB $= -0.0453$). Physical bipolar electrodes with distinct contact geometries may exhibit different spatial selectivity. Second, due to the quadratic complexity of kernel computations, the RBF-SVM was trained on a subsampled budget of 40,000 windows, rendering its performance (0.538) a conservative lower bound. Third, NinaPro DB5 utilizes a lower sampling rate (200 Hz) and 8-bit quantization, serving as an algorithmic sanity check under consumer-grade constraints rather than a clinical transfer benchmark. Future work should combine sparse physical arrays with real-time adaptive recalibration and transfer learning.

6. Conclusion

This study evaluated explainability-guided sparse sensor selection for subject-independent sEMG gesture recognition under strict zero-leakage conditions. We introduced Stability-Regularized Cross-Fitted Attribution (SR-CFA), which combines out-of-fold TreeSHAP attributions within inner training folds with between-subject dispersion penalties. Evaluated under a pre-specified non-inferiority margin ($\delta = 0.02$ Macro-F1) across 10,000 participant cluster bootstrap resamples on the 43-participant GRABMyo benchmark, pruning the 16-channel array down to $k^* = 12$ channels maintained statistical non-inferiority ($\text{LCB}_{95\%} = -0.0160 > -0.020$), achieving a 25% reduction in physical sensors.

Attribution stability analysis demonstrated consistent channel rankings across independent participants ($\rho = 0.927 \pm 0.038$), with top rankings concentrating on the primary forearm flexor and extensor muscular compartments. Translating these attributions into hardware design, a single physically fixed consensus 12-channel layout derived on development subjects generalized to unseen test participants with statistical non-inferiority ($\Delta = -0.0124$, 95% CI $[-0.0179, -0.0077]$). Longitudinal evaluations demonstrated that multi-day training exposure yielded a statistically significant improvement of +0.0210 Macro-F1 ($p < 0.001$) on subsequent sessions, confirming that multi-session training partially mitigates the spatial covariate shift induced by electrode donning and doffing.

These findings establish that explainability-guided sensor selection can identify compact, population-level electrode configu-

rations that maintain decoding fidelity across novel individuals. Replicated on NinaPro DB5, this framework provides an audited, leak-safe foundation for designing next-generation wearable myoelectric prostheses, where sparse physical electrode layouts can be synergistically paired with multi-session calibration and adaptive transfer learning.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

All raw datasets used in this study are publicly available: GRABMyo v1.1.0 via PhysioNet (`https://physionet.org/content/grabmyo/1.1.0/`) and NinaPro DB5 via the NinaPro repository. All preprocessing scripts, feature extractors, statistical analysis protocols, and manuscript figure generators are available in the open repository.

References

[1] D. Farina, R. Merletti, R. M. Enoka, The extraction of neural information from the surface emg, Journal of Applied Physiology 96 (4) (2004) 1486–1495. doi:10.1152/japplphysiol.01070.2003.

[2] D. Farina, N. Jiang, H. Rehbaum, A. Holobar, B. Graimann, H. Dietl, O. C. Aszmann, The extraction of neural information from the surface emg for the control of upper-limb prostheses: Emerging avenues and challenges, IEEE Transactions on Neural Systems and Rehabilitation Engineering 22 (4) (2014) 797–809. doi:10.1109/TNSRE.2014.2305111.

[3] I. Vujaklija, D. Farina, O. C. Aszmann, Current state and future perspective of myoelectric control: A review, IEEE Reviews in Biomedical Engineering 9 (2016) 224–249. doi:10.1109/RBME.2016.2550540.

[4] U. Cote-Allard, E. Campbell, A. Phinyomark, F. Laviolette, B. Gosselin, E. Scheme, Deep learning for electromyographic hand gesture recognition: A review, IEEE Transactions on Neural Systems and Rehabilitation Engineering 28 (12) (2020) 2901–2915. doi:10.1109/TNSRE.2020.2982810.

[5] S. Muceli, N. Jiang, D. Farina, Accurate and robust extraction of motor unit action potentials from high-density surface emg using blind source separation, IEEE Transactions on Neural Systems and Rehabilitation Engineering 22 (3) (2014) 622–633. doi:10.1109/TNSRE.2013.2282898.

[6] A. Stango, F. Negro, D. Farina, Spatial correlation and sensor layout optimization in high-density surface electromyography for myoelectric control, IEEE Transactions on Neural Systems and Rehabilitation Engineering 23 (2) (2015) 189–198. doi:10.1109/TNSRE.2014.2323352.

[7] E. Scheme, K. Englehart, Electromechanical delay and window length considerations for proportional myoelectric control, Journal of Neural Engineering 8 (3) (2011) 036005. doi:10.1088/1741-2560/8/3/036005.

[8] A. H. Al-Timemy, G. Bugmann, J. Escudero, N. Dalley, Classification of hand movements for myoelectric prostheses using emg from amputees, IEEE Transactions on Biomedical Engineering 60 (11) (2013) 3156–3165. doi:10.1109/TBME.2013.2268712.

[9] E. Campbell, A. Phinyomark, E. Scheme, Differences in the user-dependent and user-independent characteristics of surface electromyography, IEEE Transactions on Neural Systems and Rehabilitation Engineering 28 (4) (2020) 906–915. doi:10.1109/TNSRE.2020.2974955.

[10] A. Phinyomark, P. Phukpattaranont, C. Limsakul, Feature reduction and selection for emg signal classification, Expert Systems with Applications 39 (8) (2012) 7420–7431. doi:10.1016/j.eswa.2012.01.102.

[11] L. Bi, A. G. Feleke, C. Guan, High-density emg channel selection based on spatial information and mutual information for hand motion classification, Biomedical Signal Processing and Control 47 (2019) 343–352. doi:10.1016/j.bspc.2018.08.037.

[12] X. Qu, et al., Reduce surface electromyography channels for gesture recognition by multitask sparse representation and minimum redundancy maximum relevance, Journal of Healthcare Engineering 2021 (2021) 9929684. doi:10.1155/2021/9929684.

[13] A. Phinyomark, E. Scheme, Emg pattern recognition in the era of big data and deep learning, Biomedical Signal Processing and Control 42 (2018) 131–142. doi:10.1016/j.bspc.2018.01.018.

[14] S. M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 30, 2017, pp. 4765–4774.

[15] S. M. Lundberg, G. Erion, H. Chen, A. DeGrave, J. M. Prutkin, B. Nair, R. Katz, J. Himmelfarb, N. Bansal, S.-I. Lee, From local explanations to global understanding with explainable ai for trees, Nature Machine Intelligence 2 (1) (2020) 56–67. doi:10.1038/s42256-019-0138-9.

[16] G. Visani, E. Bagli, F. Chesani, Statistical stability indices for local explanations: A comprehensive empirical study, Machine Learning 111 (6) (2022) 2089–2129. doi:10.1007/s10994-022-06222-9.

[17] J. Liu, P. Zhou, Emg feature assessment for myoelectric pattern recognition and channel selection: A study with incomplete spinal cord injury, Medical Engineering & Physics 36 (8) (2014) 975–980. doi:10.1016/j.medengphy.2014.04.001.

[18] L. Hargrove, K. Englehart, B. Hudgins, The effect of electrode displacements on pattern recognition for prosthesis control, IEEE Transactions on Biomedical Engineering 54 (12) (2007) 2137–2143. doi:10.1109/TBME.2007.897827.

[19] M. M. Vidovic, H.-J. Hwang, S. Amsüss, J. M. Hahne, D. Farina, K.-R. Müller, Improving the robustness of myoelectric pattern recognition against electrode shift using high-density emg, IEEE Transactions on Neural Systems and Rehabilitation Engineering 24 (2) (2016) 258–268. doi:10.1109/TNSRE.2015.2458580.

[20] D. J. Schuirmann, A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability, Journal of Pharmacokinetics and Biopharmaceutics 15 (6) (1987) 657–680. doi:10.1007/BF01062106.

[21] E. Walker, A. S. Nowacki, Understanding equivalence and noninferiority testing, Journal of General Internal Medicine 26 (2) (2011) 192–196. doi:10.1007/s11606-010-1513-8.

[22] A. Pradhan, J. He, N. Jiang, Multi-day dataset of forearm and wrist electromyogram for hand gesture recognition and biometrics, Scientific Data 9 (1) (2022) 748. doi:10.1038/s41597-022-01836-y.

[23] S. Pizzolato, L. Tagliapietra, M. Cognolato, M. Reggiani, H. Müller, M. Atzori, Comparison of six electromyography acquisition setups on hand movement classification tasks, PLoS ONE 12 (10) (2017) e0186132. doi:10.1371/journal.pone.0186132.

[24] J. He, X. Sheng, X. Zhu, N. Jiang, Spatial information enhances multi-day gesture recognition against electrode shift, IEEE Transactions on Neural Systems and Rehabilitation Engineering 28 (10) (2020) 2208–2217. doi:10.1109/TNSRE.2020.3015433.

[25] M. Sundararajan, A. Taly, Q. Yan, Axiomatic attribution for deep networks, in: International Conference on Machine Learning (ICML), 2017, pp. 3319–3328.

[26] W. Samek, A. Binder, G. Montavon, S. Lapuschkin, K.-R. Müller, Evaluating the visualization of what a deep neural network has learned, IEEE Transactions on Neural Networks and Learning Systems 28 (11) (2016) 2660–2673. doi:10.1109/TNNLS.2016.2599820.

[27] N. Meinshausen, P. Bühlmann, Stability selection, Journal of the Royal Statistical Society: Series B (Statistical Methodology) 72 (4) (2010) 417–473. doi:10.1111/j.1467-9868.2010.00740.x.

[28] P. Kaufmann, K. Englehart, M. Platzner, Fluctuating emg signals: Investigating long-term effects of pattern recognition, in: Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), 2010, pp. 6357–6360. doi:10.1109/IEMBS.2010.5627282.

[29] S. Hooker, D. Erhan, P.-J. Kindermans, K. Been, A benchmark for interpretability methods, in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 32, 2019.

[30] G. Li, A. E. Schultz, T. A. Kuiken, Quantifying active degrees of freedom in high-density surface electromyogram for myoelectric control, IEEE

Transactions on Neural Systems and Rehabilitation Engineering 18 (5) (2010) 545–552. doi:10.1109/TNSRE.2010.2045659.